CYBER OFFENSE DETECTING AND REPORTING

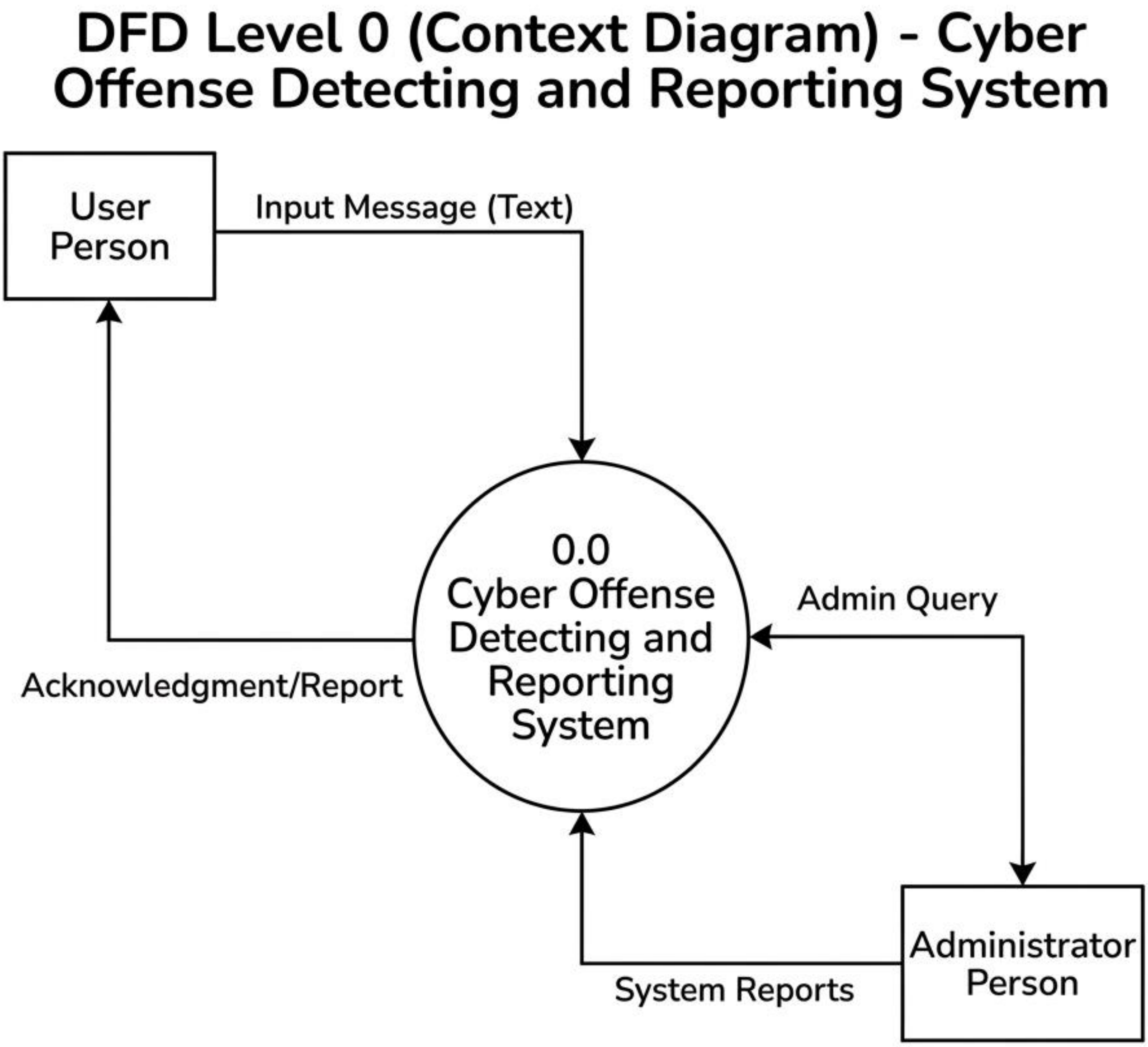
Project Documentation:

Cyber offense detecting and reporting an intelligent text analysis system designed to identify harmful, offensive, and abusive content in online communications. The system analyzes user-generated text from social media platforms, messages, comments, and online forums to determine whether the content contains cyberbullying behavior. By utilizing Natural Language Processing (NLP) the system classifies text as bullying or non-bullying. The project helps social media platforms, organizations, and communities create a safer online environment by automatically detecting and reporting harmful content.

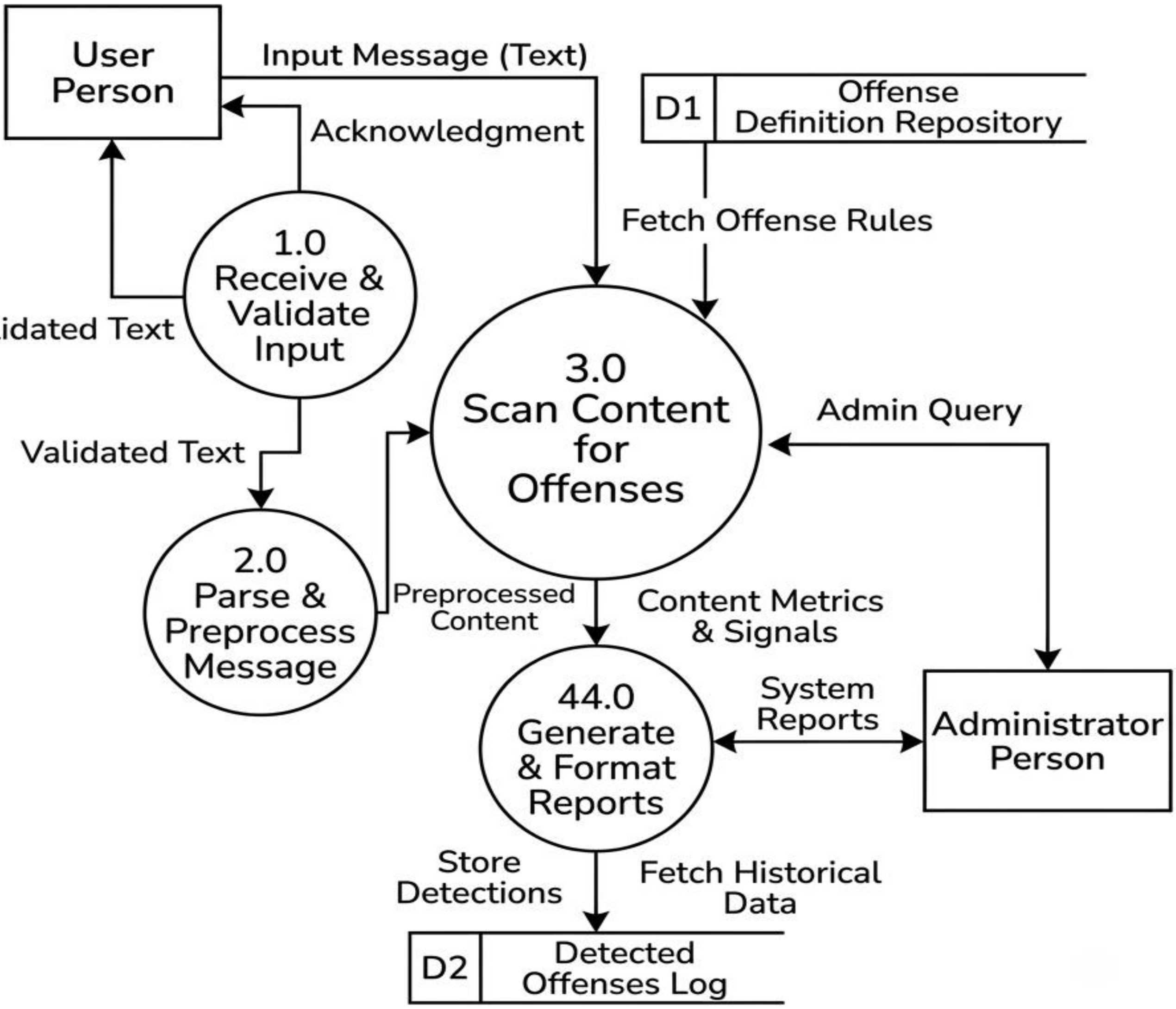
Project Overview:

The system begins when a user enters or uploads textual content for analysis. The input text is first validated and processed through preprocessing stages such as tokenization, stop-word removal, punctuation removal, and text normalization. After preprocessing, important textual features are extracted using techniques such as TF-IDF. The processed data is then analyzed using analysis to determine the emotional tone of the message. Subsequently, machine learning algorithms classify the content as cyberbullying or non-cyberbullying based on learned patterns from training datasets. The final prediction result, along with information, is displayed to the user. Detection history, classification results, and user activities can be stored in the database for future analysis and reporting purposes.

Data Flow Diagrams:



DFD LEVEL 1 DIAGRAM: CYBER OFFENSE DETECTING AND REPORTING SYSTEM



Database Table Design:

1. User Table

Field Name	Data Type	Key	Description
user_id	INT	PK	Unique User ID
username	VARCHAR(100)	-	User Name
email	VARCHAR(150)	-	Email Address
password	VARCHAR(255)	-	Encrypted Password
created_at	DATETIME	-	Account Creation Date

2. REPORTS

Field Name	Data Type	Key	Description
report_id	INT	PK	Unique Report ID
user_id	INT	FK	User Reference
platform	VARCHAR(50)	-	Social Media Platform
report_type	VARCHAR(100)	-	Cyber Offense Category

Field Name	Data Type	Key	Description
description	TEXT	-	Complaint Details
report_date	DATETIME	-	Date of Report

3. SOCIAL_MEDIA_POSTS

Field Name	Data Type	Key	Description
post_id	INT	PK	Unique Post ID
user_id	INT	FK	User Reference
post_content	TEXT	-	Social Media Content
platform	VARCHAR(50)	-	Facebook, Twitter, YouTube
created_at	DATETIME	-	Post Creation Time

4. OFFENSE_CLASSIFICATION

Field Name	Data Type	Key	Description
classification_id	INT	PK	Classification ID
post_id	INT	FK	Post Reference
offense_type	VARCHAR(100)	-	Harassment, Abuse, Threat
confidence_score	FLOAT	-	Prediction Confidence
status	VARCHAR(20)	-	Offensive / Non-Offensive

Modules:

User Authentication Module:

This module manages secure login and access control for users and administrators. It ensures that only authorized users can access the system and its functionalities.

Text Input Module:

This module allows users to enter or upload textual content such as social media comments, messages, posts, or chat conversations for cyberbullying analysis.

Text Preprocessing Module:

This module cleans and prepares the input text by performing operations such as tokenization, stop-word removal, punctuation removal, lowercasing, and text normalization to improve data quality.

Feature Extraction Module:

This module converts the processed text into numerical features using techniques such as TF-IDF (Term Frequency-Inverse Document Frequency), making the data suitable for machine learning algorithms.

Machine Learning Classification Module:

This module applies machine learning algorithms such as Logistic Regression, Naïve Bayes, or Support Vector Machine (SVM) to classify the text as Cyberbullying or Non-Cyberbullying.

Result Display Module:

This module presents the prediction result to the user, showing whether the input text contains cyberbullying information and confidence score.

Report Generation Module:

This module generates analysis reports containing classification results, prediction statistics, and downloadable summaries for future reference.

Admin and History Management Module:

This module enables administrators to monitor user activities, manage system records, review detection history, and maintain overall system performance and security.

Project Screenshots:

DASHBOARD PAGE

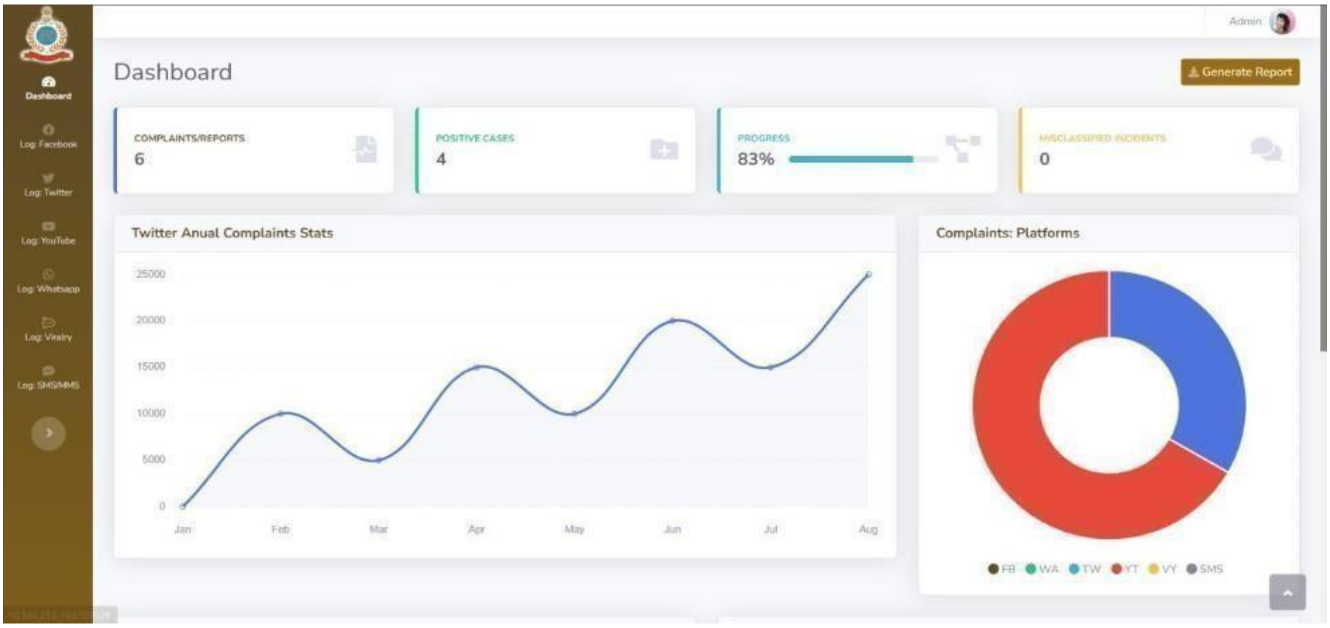


Fig .Dashboard Page

4.1 REPORTING PLATFORM

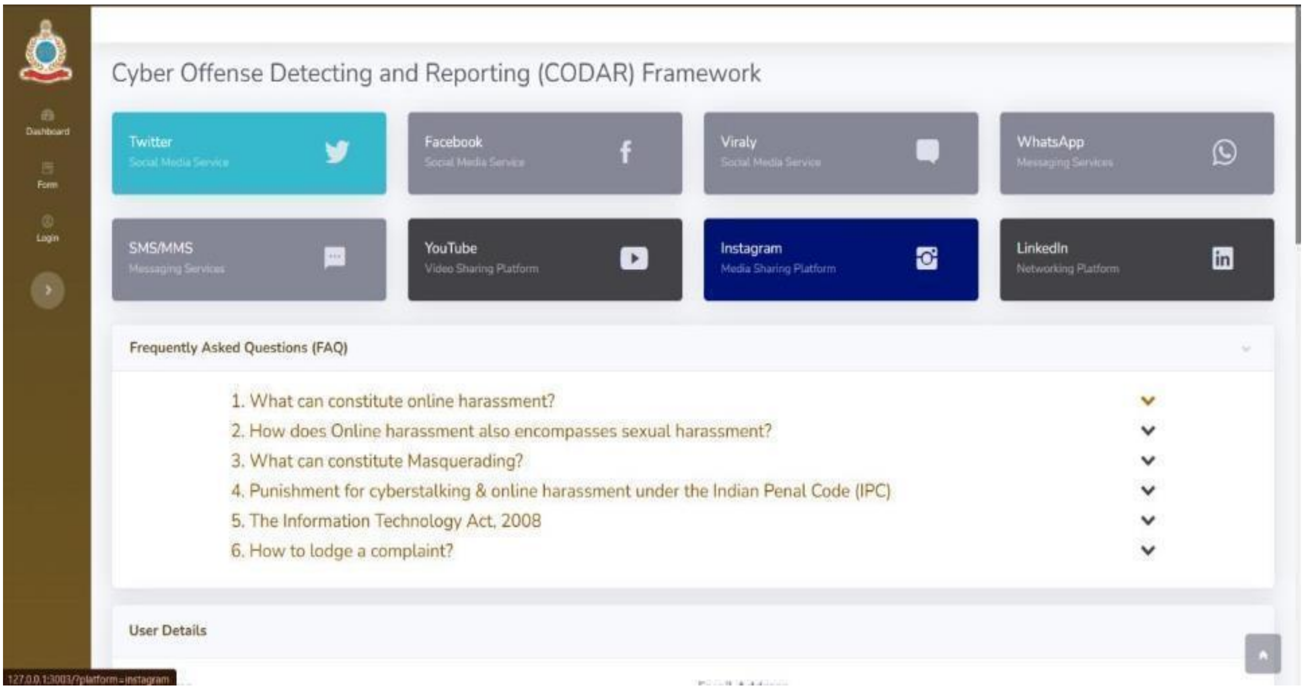


Fig Reporting Platform

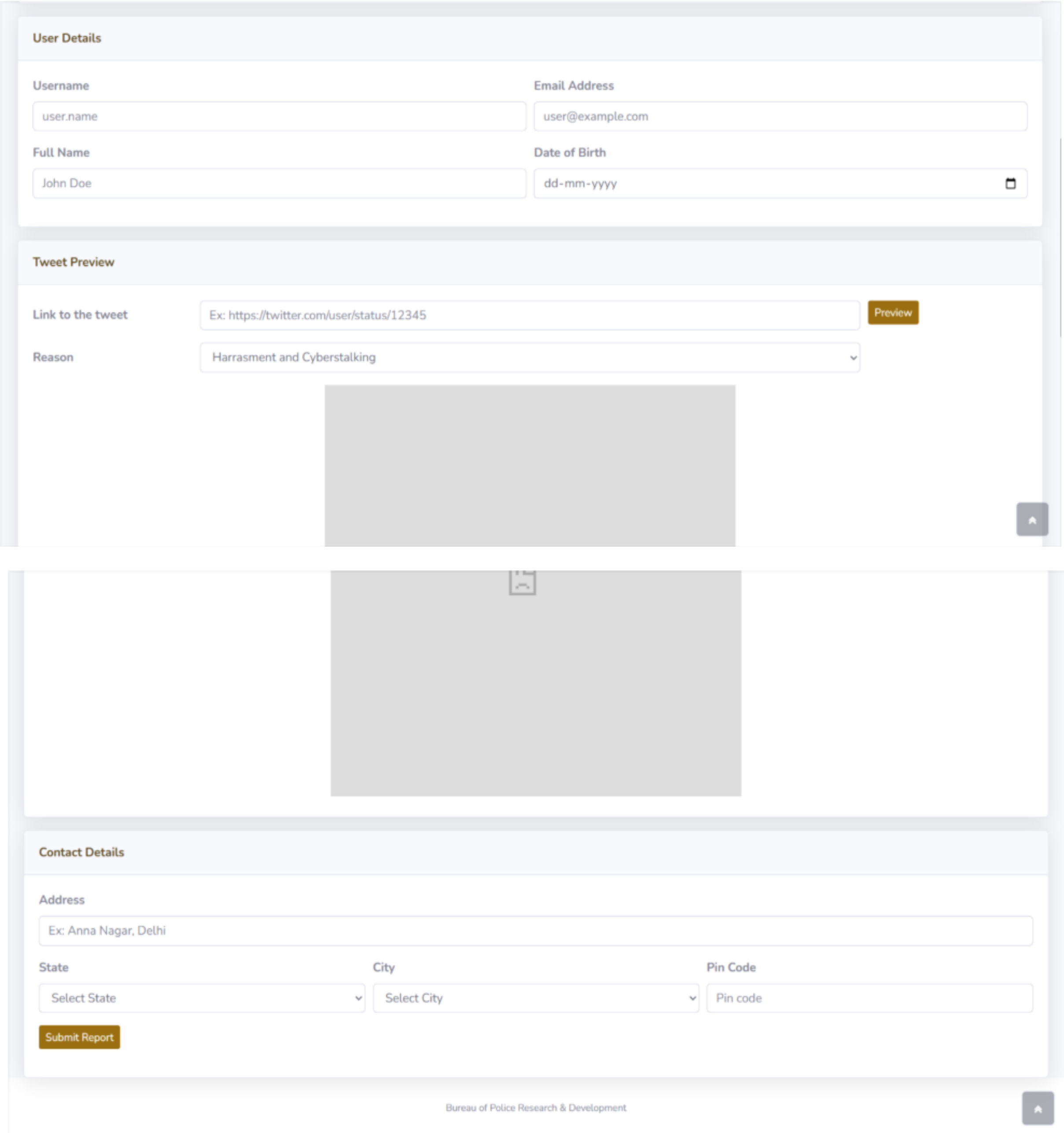


Fig Reporting Platform

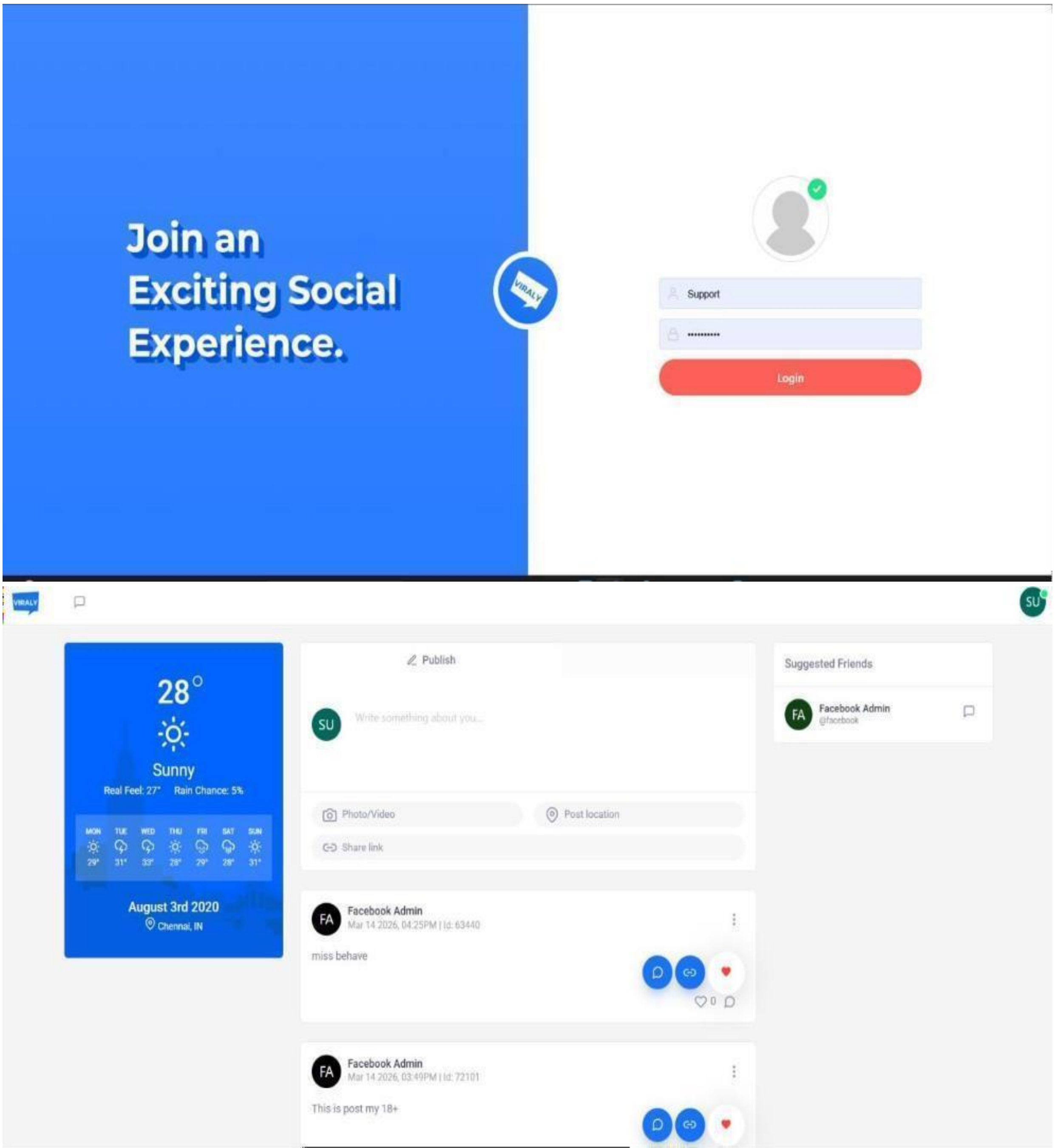
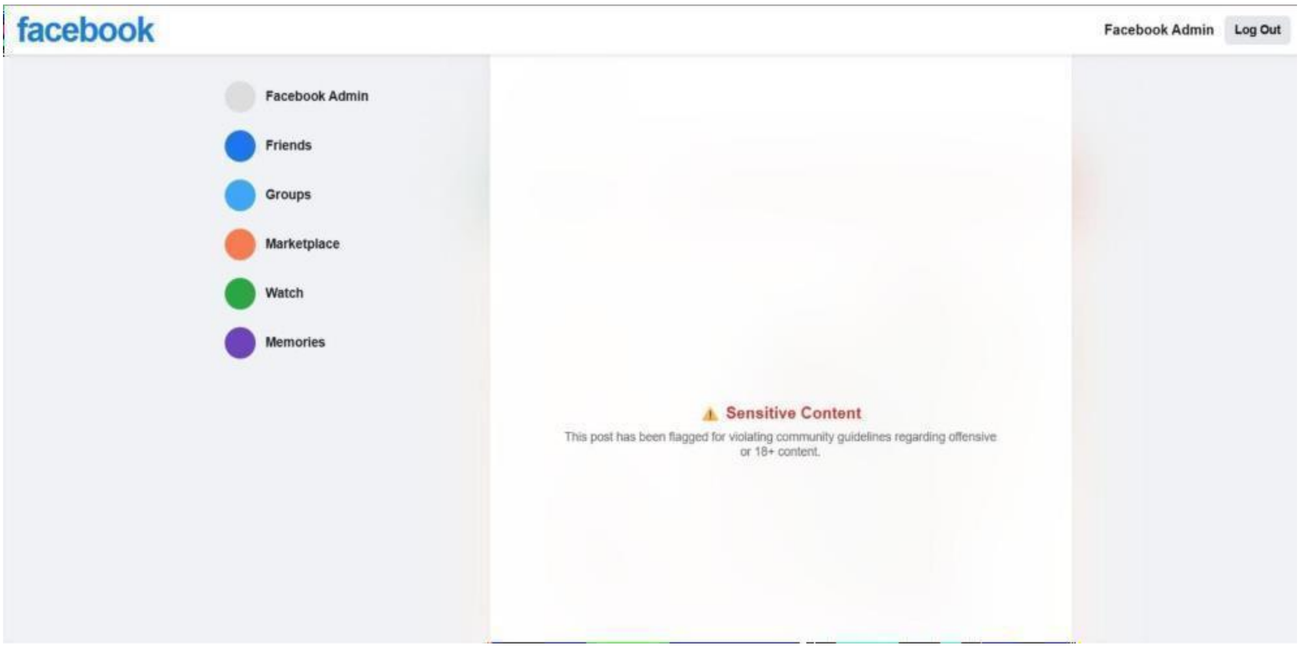
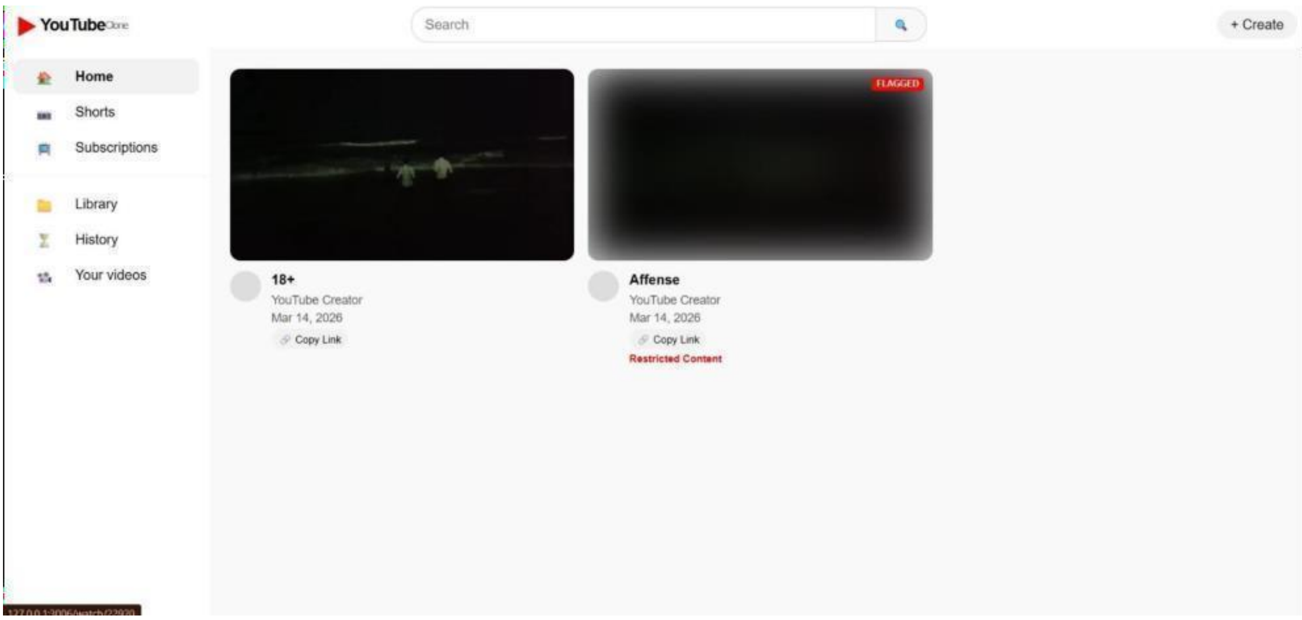


Fig Social Media Platform



Fig, FaceBook Platform



Fig,Youtube Platform